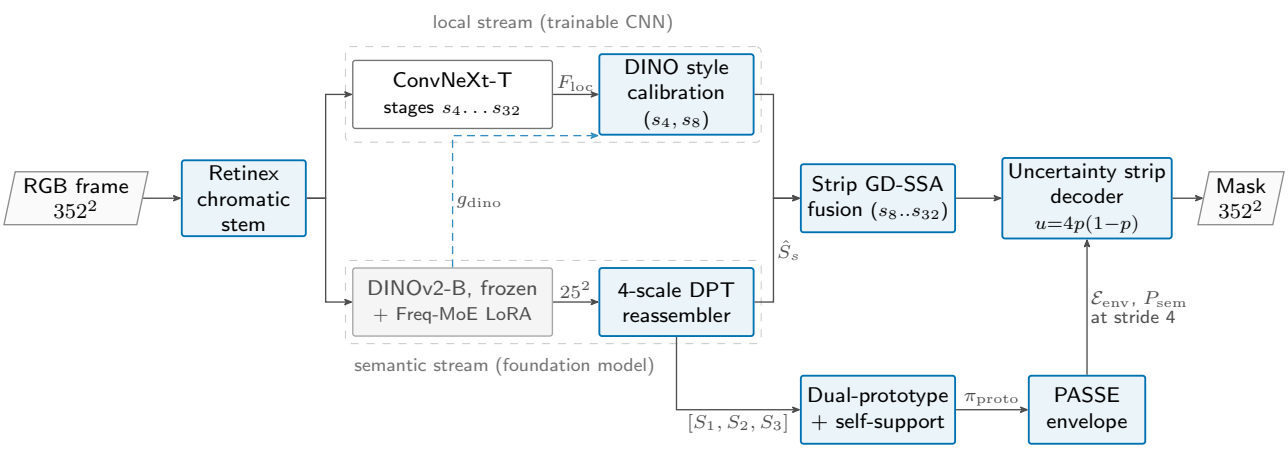

local stream (trainable CNN)
RGB frame
352^2
Retinex
chromatic
stem
ConvNeXt-T
stages $s_4 \ldots s_{32}$
F_{loc}
DINO style
calibration
(s_4, s_8)
g_{dino}
DINOv2-B, frozen
+ Freq-MoE LoRA
25^2
4-scale DPT
reassembler
$\hat{S}_s$
semantic stream (foundation model)
Strip GD-SSA
fusion $(s_8..s_{32})$
Uncertainty strip
decoder
$u{=}4p(1{-}p)$
Mask
352^2
$\mathcal{E}_{\text{env}}$, P_{sem}
at stride 4
PASSE
envelope
π_{proto}
Dual-prototype
+ self-support
$[S_1, S_2, S_3]$